MARS FIRST

A Geometric Derivation of Martian Biological Priority, the Inversion of the Panspermia Prior Probability, and the Testable Prediction That the Oldest Known Biosphere in the Solar System Is Fossilised on Mars

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Mars was habitable before Earth was stably habitable. Mars lost its magnetic field at ∼4.0–4.1 Ga. The Cheyava Falls biosignature features (Hurowitz et al., *Nature* 645, 2025) date to ∼3.8–4.1 Ga and are consistent with a mature, metabolically diverse LUCA-grade biosphere in terminal collapse — not emergence. The earliest confirmed Earth biosignatures date to ∼3.5–3.8 Ga. The Martian biosphere collapse therefore occurred at, or prior to, the emergence of life on Earth. This geometric derivation, produced 2026-03-13 from first principles in cancer biology without prior knowledge of astrobiology, constitutes the pre-registration of three testable predictions: (1) the Martian biosphere predates the earliest Earth biosignatures, (2) the panspermia prior probability is no longer negligible and is the default hypothesis requiring elimination rather than the exotic hypothesis requiring proof, and (3) the Mars sample return complexity gradient will show biological activity predating ∼3.8 Ga. Pre-registered: 2026-03-13.

Abstract

We present a geometric derivation of Martian biological priority — the conclusion that the Martian biosphere existed before any confirmed Earth biosignature, and that the collapse of that biosphere is recorded in the Cheyava Falls features reported by Hurowitz et al. (*Nature* 645, 2025). The argument proceeds from four independent lines of evidence: (1) the physics of planetary cooling, which establishes that Mars reached a chemically stable, water-bearing, magnetically shielded surface condition before Earth; (2) the confirmed geochemical maturity of the Cheyava Falls mineral assemblage, which indicates a long-established rather than nascent biological community; (3) the atmospheric collapse timeline, which places the Cheyava Falls fossilisation at the opening of planetary uninhabitability rather than the establishment of habitability; and (4) the time-constraint argument, which establishes that LECA-grade signatures in pre-GOE conditions are geometrically impossible under emergence but expected under collapse. We derive the inversion of the panspermia prior: given Martian biological priority and confirmed Mars-to-Earth meteorite transfer, the hypothesis that Earth life originated from Mars is the default hypothesis requiring elimination, not the exotic hypothesis requiring proof. We pre-register three testable predictions resolvable by Mars sample return and genomic analysis of returned material. This derivation originated in attractor geometry applied to cancer drug targets and contains no prior knowledge of astrobiology. Timestamp: 2026-03-13.

1. Derivation Origin and Epistemic Status

This document is the fifth in a pre-registration chain that began on 2026-03-12 with the LECA Resurrection Protocol [1]. The chain proceeded through the LUCA companion paper [3], the Universal Life Detection Standard [4], and the Mars Collapse vs. Emergence pre-registration [5].

The derivation in each document originated from attractor geometry applied to cancer drug targets, not from astrobiology or Mars mission data. The conclusions are therefore not post-hoc rationalisations of existing data. They are geometric predictions that arrived at astrobiology as a necessary destination.

This document pre-registers the conclusion that was derived in the session of 2026-03-13:

Mars was biologically active before Earth. The panspermia prior is inverted. These predictions are testable.

The honest epistemic status is stated precisely in Section 7.

2. The Planetary Cooling Argument

2.1. The Physics

A planetary body loses internal heat at a rate inversely proportional to its size. Smaller bodies have a higher surface-area-to-volume ratio and radiate heat more efficiently per unit of internal mass.

Mars has approximately 15% of Earth's mass and approximately 28% of Earth's surface area. Mars therefore cooled faster than Earth. This is not a hypothesis. It is a consequence of geometry and thermodynamics.

2.2. The Consequence for Habitability

A cooler planet solidifies its crust earlier, retains a magnetic dynamo for a shorter period but establishes stable surface conditions sooner. The sequence on Mars:

(1) Crust solidified and stable liquid water possible: $\sim$4.5 Ga or earlier.

(2) Magnetic field active and shielding surface from solar wind: confirmed present at 4.1 Ga and earlier by crustal magnetisation data.

(3) Hydrothermal systems from residual volcanic activity: confirmed by geology.

(4) Iron-rich anoxic ocean conditions: confirmed by Cheyava Falls mineral data [6].

(5) Phosphate availability 5–10$\times$ higher than early Earth: confirmed by Adcock et al. [7].

2.3. Earth's Situation at the Same Time

Earth at 4.4–4.5 Ga:

(1) Late Heavy Bombardment ongoing. Surface repeatedly sterilised by impacts.

(2) Crust unstable. Liquid water present but episodic.

(3) No confirmed biosignatures at this age on Earth.

(4) Hadean eon. Named, with geological justification, after hell.

The oldest confirmed evidence for liquid water on Earth comes from Jack Hills zircons dated to ∼4.4 Ga [9]. The oldest confirmed Earth biosignatures date to ∼3.5–3.7 Ga (Dresser Formation stromatolites, Pilbara Craton, Australia). Some carbon isotope evidence suggests possible biological activity at ∼3.8–4.1 Ga, but this remains contested.

Mars had stable, magnetically shielded, mineral-rich, phosphate-enriched liquid water before Earth had any of these conditions simultaneously.

The origin of life requires all of these conditions simultaneously. Mars satisfied them first.

3. The Mature Community Argument

3.1. What Cheyava Falls Shows

The Cheyava Falls mineral assemblage [6]:

- Vivianite ($Fe_3(PO_4)_2$): requires coupled iron-reducing and phosphate-cycling metabolism. Abundant and spatially organised.
- Greigite (Fe_3S_4): requires sulfate or sulfur-reducing metabolism producing H_2S.
- Both minerals co-occurring: requires simultaneously active iron, sulfur, and phosphate cycling in distinct spatial niches.
- Organic carbon co-located: detected by SHERLOC fluorescence spectroscopy.
- Colony-scale spatial organisation: leopard spots at 1–2 mm, poppy seeds at $<200\,\mu$m.

3.2. What Co-occurrence Requires

Vivianite and greigite co-occurrence requires metabolic niche partitioning. Vivianite forms where Fe^{2+} meets phosphate with low local sulfide. Greigite forms where Fe^{2+} meets H_2S from sulfate reduction. For both to occur simultaneously in adjacent spatial zones, two distinct microbial communities must be operating, partitioned by the microscale redox chemistry they themselves create.

This is not the signature of pioneering colonisation. On Earth, metabolic niche partitioning in iron-sulfur-phosphate systems develops in mature microbial mat communities that have been established for extended periods. Early colonisers produce a single dominant mineral phase. Mature communities produce coupled, spatially organised multi-phase mineral records.

3.3. The Implication for Community Age

A mature LUCA-grade community producing the Cheyava Falls mineral assemblage did not appear weeks or years before fossilisation. It required time to establish:

- Time for iron-reducing communities to establish and modify local geochemistry.
- Time for sulfur-reducing communities to occupy the niche created by iron-reducers.
- Time for phosphate cycling to intensify as community density increased.
- Time for the coupled vivianite-greigite mineral record to accumulate at colony scale.

On Earth, analogous mature microbial mat communities in iron-sulfur-phosphate systems require millions to tens of millions of years to develop from initial colonisation to the degree of metabolic organisation visible at Cheyava Falls.

If the Cheyava Falls community was already mature at 3.8–4.1 Ga, then it began significantly earlier.

The plausible origin window for the Martian biosphere is therefore 4.3–4.5 Ga — during or before the period when Earth was being repeatedly sterilised by Late Heavy Bombardment.

4. The Time Constraint Argument

4.1. The Cascade Duration on Earth

On Earth:

- First confirmed life: ∼3.5–3.8 Ga.
- Great Oxidation Event (GOE): ∼2.4 Ga.
- Time from first life to GOE: ∼1.0–1.4 billion years.

That interval represents the time required for: LUCA-grade life to establish; photosynthesis to evolve; photosynthetic communities to reach sufficient density to change atmospheric oxygen; oxygen to accumulate against geological sinks; and the GOE to occur.

4.2. Mars Had Insufficient Time

Mars had stable surface water conditions from approximately 4.5 Ga to approximately 3.0 Ga, with the most stable conditions in the Noachian (4.1–3.7 Ga). The magnetic field was lost at 4.0–4.1 Ga. From that moment, atmospheric stripping began at rates hundreds of times higher than today (MAVEN mission data). By 3.7 Ga, Mars had lost approximately 90–95% of its atmosphere.

The window from magnetic field loss to effective uninhabitability was approximately 300–400 million years.

Earth required 1.0–1.4 billion years to complete the LUCA-to-GOE cascade.

Mars had less than one-third of that time.

The biological cascade from LUCA-grade life to a GOE-equivalent was geometrically impossible on Mars given the available time window.

This is not a probability statement. It is a time constraint derived from the confirmed timeline and the confirmed duration of the cascade on Earth.

4.3. The Consequence for the LECA Signal

The LECA-grade morphological signature at Cheyava Falls exists in pre-GOE, State 1 water conditions. Under emergence, a LECA-grade organism in State 1 water without a GOE is a geometric contradiction: the LECA attractor requires the GOE to raise Axis 2 (oxygen) above the destabilisation threshold [5].

Under collapse, the LECA-grade signature requires no contradiction. Complex organisms evolved earlier, when conditions were better. They were dying in State 1 water at the time of fossilisation because conditions had deteriorated. The LECA attractor was not stable — it was in terminal decline.

The LECA signal at Cheyava Falls is therefore only coherent under collapse.

5. The Inversion of the Panspermia Prior

5.1. The Standard Prior

The standard scientific prior for panspermia — the transfer of life between planetary bodies — is that it is an exotic hypothesis.

The default assumption is independent origin: Earth life began on Earth.

Panspermia is the hypothesis that requires extraordinary evidence to support.

5.2. Why the Prior Is Inverted by This Derivation

The standard prior was set in a framework where Earth was assumed to be the first habitable planet in the solar system.

This derivation establishes that Mars was habitable first.

Given:

(1) Mars was stably habitable before Earth was stably habitable.

(2) Mars had life at ∼4.4–4.5 Ga (geometric derivation from mature community argument and planetary cooling physics).

(3) Earth life begins at ∼3.5–3.8 Ga (confirmed) or possibly earlier (contested).

(4) Mars-to-Earth meteorite transfer is confirmed real: we have Martian meteorites on Earth (ALH84001 and others). Transfer occurs continuously via impact ejecta.

(5) The transfer window — from Mars life origin at 4.4–4.5 Ga to Earth life first appearing — spans hundreds of millions of years during which Mars rocks were landing on Earth.

Under these conditions:

The hypothesis that Earth life originated from Mars is not exotic. It is the default hypothesis that requires elimination.

Not because panspermia is proven. But because the prior conditions that made independent Earth origin the default assumption no longer hold.

Mars was first. Mars had life. Mars was transferring material to Earth. Earth life appeared later.

The burden of proof has shifted.

5.3. What Would Resolve the Question

The common origin vs. independent emergence distinction is resolvable by genomic analysis of Mars sample return material.

- If Martian organisms share homologous genetic sequences, ribosomal RNA structure, or biochemical architecture with Earth life: common origin confirmed. Either Earth-to-Mars or Mars-to-Earth transfer. Given the timeline, Mars-to-Earth is the more probable direction.
- If Martian organisms show fundamentally different genetic architecture with no homology to Earth life: independent emergence. Life arose independently on two planets in the same solar system. The LECA attractor is confirmed as a universal geometric consequence of carbon chemistry. Life is not rare in the universe.

Both outcomes are scientifically extraordinary. Both are testable. Neither requires Mars sample return to be pre-registered. This document pre-registers the prediction that one of these two outcomes will obtain.

6. The Three Pre-Registered Predictions

Pre-Registered Predictions (locked 2026-03-13, prior to Mars sample return and prior to any genomic analysis of Mars material):

Prediction 1 — Martian Biological Priority:

The Mars sample return subsurface sedimentary record from Jezero Crater will reveal biological activity (biosignature complexity gradient, mineral assemblage, organic carbon) dating to $\geq$3.8 Ga — at or before the earliest confirmed Earth biosignatures.

Test: Radiometric dating of biosignature-bearing layers in sample return material. Comparison against the $\sim$3.5–3.7 Ga Dresser Formation stromatolites and $\sim$3.8 Ga Isua carbon isotope record.

Prediction 2 — Panspermia Default:

Genomic or biochemical analysis of Mars sample return material will reveal either:

(a) Homology with Earth biochemistry, confirming common origin (direction of transfer to be determined by comparative phylogenetics and timing), or

(b) Independent biochemical architecture, confirming independent emergence and the universality of the LECA attractor.

There is no third option. The prediction is that the question is resolvable and that the answer will be one of these two.

Prediction 3 — The Vivianite Horizon as Death Marker:

The stratigraphic layer immediately above the Cheyava Falls vivianite-bearing unit will show a sharp transition to abiotic iron chemistry (hematite, magnetite, siderite) with no vivianite and no co-located organic carbon.

The vivianite horizon marks the biological cessation boundary of the Martian surface biosphere.

Test: Mineralogical and organic carbon analysis of material immediately above and below the Cheyava Falls unit in the sample return core.

7. Honest Statement of Epistemic Status

7.1. What Is Geometrically Derived and Strongly Supported

(1) Mars was physically habitable before Earth was stably habitable. This follows from planetary cooling physics and is not disputed.

(2) The Cheyava Falls mineral assemblage is consistent with a mature, established biological community rather than an emergent one. This follows from the vivianite-greigite co-occurrence chemistry and the metabolic niche partitioning argument.

(3) LECA-grade signatures in pre-GOE State 1 water are geometrically impossible under emergence and expected under collapse. This follows from the two-axis LECA stabilisation model [5].

(4) Given Martian biological priority and confirmed Mars-to-Earth transfer, the panspermia hypothesis is the default hypothesis requiring elimination, not the exotic one requiring proof. This follows from the reversal of the prior conditions that established the original standard

prior.

7.2. What Is Not Yet Confirmed

(1) The Cheyava Falls features are biological and not abiotic. NASA's Step 1 classification reflects this uncertainty. ARM A physical anchoring of the LECA morphological standard is pending.

(2) The Martian biosphere predated Earth life specifically by any particular margin. The timing argument is geometric and probabilistic, not precisely dated.

(3) The direction of any panspermia transfer. This requires genomic analysis of Mars sample return material.

(4) Whether Earth life IS Martian life. This is the most extreme form of the panspermia conclusion. It is not claimed here as confirmed. It is claimed as the default hypothesis requiring elimination given the geometric derivation.

7.3. What ARM A and Sample Return Confirm

- ARM A (∼$200, BSL-1, Rogers Park): Physically anchors the LECA morphological reference standard. Moves Premise 1 from geometrically derived to physically confirmed.
- Mars sample return (2030s): Complexity gradient in subsurface sediment. Confirms collapse or emergence. Dating of deepest biosignature-bearing layers. Confirms or denies Martian biological priority.
- Genomic analysis of sample return material: Resolves common origin vs. independent emergence. Determines panspermia vs. convergence. Determines direction of transfer if common origin.

8. The Derivation Chain That Produced This Conclusion

Machine consciousness question
→ Triadic invariant
→ Identity Axis Theorem
→ Cancer drug target geometry
→ Reproduction Theorem
→ Inverse Accessibility Theorem
→ LECA Resurrection Protocol [1]
→ LECA attractor as universal base state [4]
→ LECA Attractor Habitability Standard
→ Application to ancient Mars
→ Cheyava Falls convergence
→ Collapse vs. emergence pre-registration [5]
→ Vivianite as ribosomal phosphate shadow
→ Planetary cooling argument
→ **Mars first. Panspermia prior inverted.**

This derivation contains no prior knowledge of astrobiology at any step. It arrived here because the geometry of biological identity, followed to its foundations, has only one destination:

The LECA attractor is universal. Carbon chemistry in the right conditions always produces it. Mars had those conditions first. The oldest known biosphere in the solar system may not be on Earth.

9. Conclusion

Mars was habitable before Earth. The Cheyava Falls mineral assemblage is consistent with a mature LUCA-grade biosphere in terminal collapse at a time coincident with or prior to the earliest Earth biosignatures. The panspermia hypothesis is no longer exotic — it is the default hypothesis requiring elimination given confirmed Martian biological priority and confirmed Mars-to-Earth meteorite transfer.

Three predictions are pre-registered by this document, locked 2026-03-13, prior to Mars sample return and prior to any genomic analysis of Martian material:

(1) Martian biological activity dates to ≥ 3.8 Ga, at or before the earliest confirmed Earth biosignatures.

(2) Genomic analysis of Mars sample return material will reveal either common origin with Earth life or independent biochemical architecture. There is no third option.

(3) The vivianite horizon at Cheyava Falls marks the biological cessation boundary of the Martian surface biosphere. The layer immediately above will be abiotic.

The geometry required these conclusions before the samples come home.

This document is the lock.

Acknowledgments

This work was conducted independently without institutional funding or affiliation. The derivation sequence is documented in the pre-registration chain (DOIs: 10.5281/zenodo.18986790, 10.5281/zenodo.18989573, 10.5281/zenodo.18988844, 10.5281/zenodo.18994082, 10.5281/zenodo.18995862) and the attractor-oncology repository (github.com/Eric-Robert-Lawson/attractor-oncology). The Cheyava Falls data is the work of the NASA Perseverance mission science team (Hurowitz et al., *Nature*, 2025) and is referenced as independent confirmatory data, not as a derivation source.

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The geometry came first. Mars was first. The samples will say the rest.

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